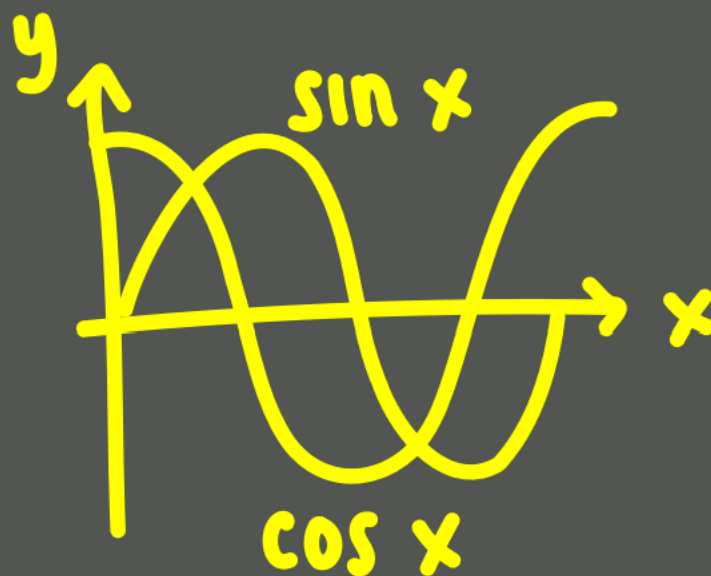


Class - 12

Trigonometry



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↓

Formula Sheet



By - Mandeep
CC : Green Board

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

Trigonometric Ratios

$$(a) \sin\theta = \frac{1}{\operatorname{cosec}\theta}$$

$$(b) \tan\theta = \frac{1}{\cot\theta}$$

$$(c) \cos\theta = \frac{1}{\sec\theta}$$

$$(d) \cot\theta = \frac{\cos\theta}{\sin\theta}$$

$$(e) \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Trigonometric Identities

$$(a) \sin^2\theta + \cos^2\theta = 1$$

$$(b) \sec^2\theta = 1 + \tan^2\theta$$

$$(c) \operatorname{cosec}^2\theta = 1 + \cot^2\theta$$

Degree to Radian Measure :

Degree	30°	45°	60°	90°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Trigo Table

Angle (°)	0°	30°	45°	60°	90°
sin x	0	1/2	1/√2	√3/2	1
cos x	1	√3/2	1/√2	1/2	0
tan x	0	1/√3	1	√3	∞
cot x	∞	√3	1	1/√3	0
cosec x	∞	2	√2	2/√3	1
sec x	1	2/√3	√2	2	∞

Formulas & Identities :

$$\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$$

$$\operatorname{cosec}^{-1} \frac{1}{x} = \sec^{-1} x$$

$$\tan^{-1} \frac{1}{x} = \cot^{-1} x$$

$$\sin^{-1}(-x) = -\sin^{-1} x$$

$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

$$\tan^{-1}(-x) = -\tan^{-1} x$$

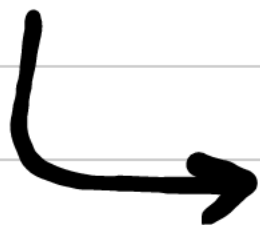
$$\cos^{-1}(-x) = \pi - \cos^{-1} x$$

$$\sec^{-1}(-x) = \pi - \sec^{-1} x$$

$$\cot^{-1}(-x) = \pi - \cot^{-1} x$$

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Trigonometric
functions



	Domain	Range
$\sin x$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	$[-1, 1]$
$\operatorname{cosec} x$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$	$\mathbb{R} - (-1, 1)$
$\tan x$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	$\mathbb{R}$
$\cos x$	$[0, \pi]$	$[-1, 1]$
$\sec x$	$[0, \pi] - \{\frac{\pi}{2}\}$	$\mathbb{R} - (-1, 1)$
$\cot x$	$(0, \pi)$	$\mathbb{R}$

	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$

Inverse
Trigonometric
functions



Addition/Subtraction Formulae

$$(a) \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$(b) \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$(c) \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$(d) \cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$(e) \cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

$$(f) \sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$(g) \tan(A + B) = \frac{(\tan A + \tan B)}{(1 - \tan A \tan B)}$$

$$(h) \tan(A - B) = \frac{(\tan A - \tan B)}{(1 + \tan A \tan B)}$$

$$(i) \cot(A + B) = \frac{(\cot B \cot A - 1)}{(\cot B + \cot A)}$$

$$(j) \cot(A - B) = \frac{(\cot B \cot A + 1)}{(\cot B - \cot A)}$$

More Formulas for Angles

(a) $\sin 2A = 2 \sin A \cos A$

(b) $\sin A = 2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right)$

(c) $\cos 2A = \cos^2 A - \sin^2 A$

(d) $\cos A = \cos^2\left(\frac{A}{2}\right) - \sin^2\left(\frac{A}{2}\right)$

(e) $\cos 2A = 2 \cos^2 A - 1$

(f) $2 \cos^2 A = 1 + \cos 2A$

(g) $\cos 2A = 1 - 2 \sin^2 A$

(h) $2 \sin^2 A = 1 - \cos 2A$

(i) $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$

(j) $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$

(k) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

(l) $\sin 3A = 3 \sin A - 4 \sin^3 A$

(k) $\cos 3A = 4 \cos^3 A - 3 \cos A$

(l) $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$

Sum to Product Identities

$$(a) \sin C + \sin D = 2 \sin \left(\frac{C + D}{2} \right) \cos \left(\frac{C - D}{2} \right)$$

$$(b) \sin C - \sin D = 2 \cos \left(\frac{C + D}{2} \right) \sin \left(\frac{C - D}{2} \right)$$

$$(c) \cos C + \cos D = 2 \cos \left(\frac{C + D}{2} \right) \cos \left(\frac{C - D}{2} \right)$$

$$(d) \cos C - \cos D = -2 \sin \left(\frac{C + D}{2} \right) \sin \left(\frac{C - D}{2} \right)$$

$$(e) 2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$(f) 2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$(g) 2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$(h) 2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

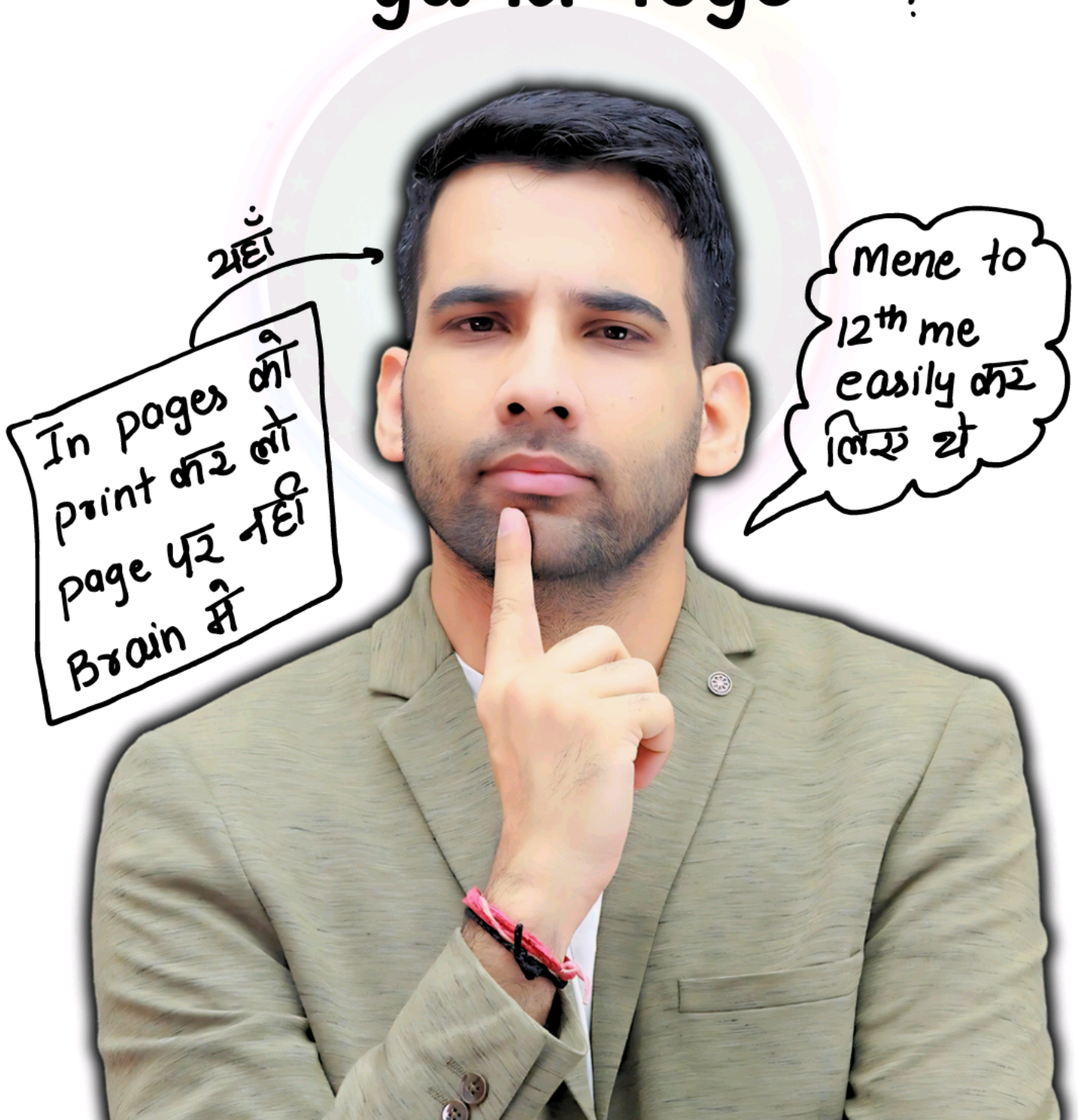


रुको अभी आगे पेज पर और भी है



Are bs itne hi the

Drr.. gye formula dekh kr
ya kr loge..... ?



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Bye

